

Furniture Market Positioning Analyzer

Predicting Consumer Interest in Furniture Products Using IKEA Datas

STAT 418 Final Project Proposal · Shiyi He

Project Overview & Motivation

Problem Statement

Furniture brands offer large product catalogs, but it is unclear which product attributes drive consumer interest.

Goal

Predict whether a furniture product will receive high consumer interest using product-level features.

Motivation

Helps brands improve pricing and product positioning strategies.

Impact

Better prediction of high-interest products can help brands reduce risk in product launches and allocate marketing resources more effectively.

Product Data

IKEA Pilot Dataset

Price Analysis

Distribution & Category Patterns

Interest Prediction

Binary Classification

Target Users: Furniture retailers, product managers, market analysts.

Data Collection Approach & Progress

Data Source

IKEA US website —Category pages + product detail pages
Bookcases, desks, sofas, beds

Collection Method

- Current: Requests + BeautifulSoup
 - Scrape product listing pages
 - Extract product URL, name, category, price
- Planned: Product detail page scraper
 - Collect color, material, dimensions, description
- Advanced Method: Playwright or XHR inspection
 - handle JS-rendered ratings and reviews if needed

Collected Fields

Now:

Product_name
Category
Price
product_url

Planned:

Color
Material
Dimensions
Description
Rating
review_count

category	price	name	high_interest
bookcases	79	BILLY Bookcase	1
bookcases	49	BILLY Bookcase	1
bookcases	59	New BILLY Book	1
bookcases	149	BRIMNES Book	0
bookcases	160	BILLY Bookcase	0
bookcases	32.99	BAGGEBO Book	0
bookcases	178	BILLY Bookcase	0
tables/desks	249.99	New MITTPLAN	1
tables/desks	199.99	CENTERHALV C	1
tables/desks	299.99	BOLLSIDAN De	1
tables/desks	329.99	MARKUS Office	1

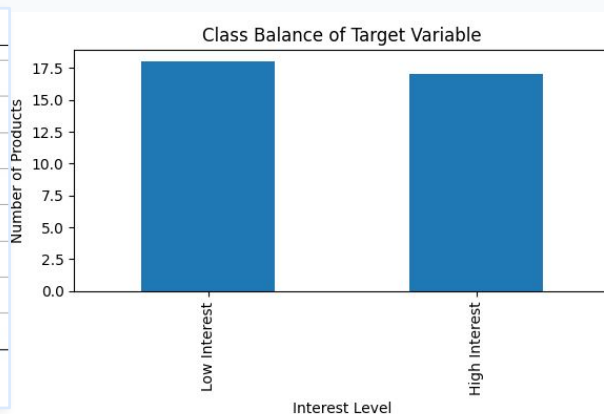
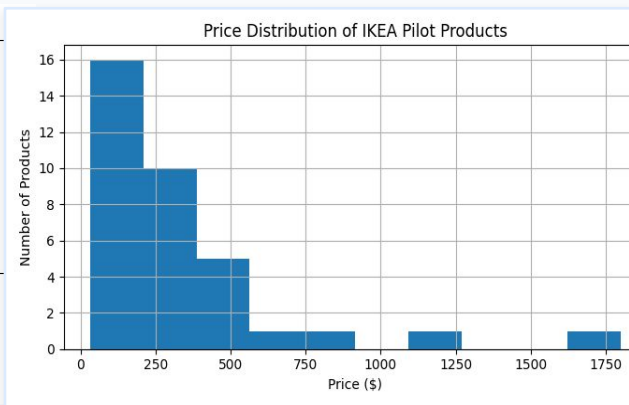
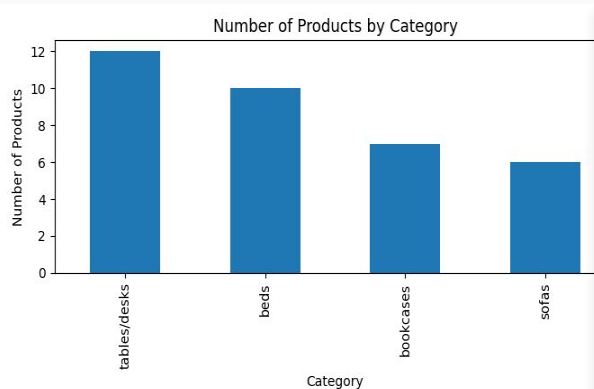
 **Key Challenge:** Product Pages - duplicate variants and some fields are dynamically rendered.

- ✓ Current Solution: Filter valid price rows and remove duplicates by product_url.
- ✓ Planned Solution: Use product detail pages and inspect XHR calls or Playwright if ratings/reviews are not available in static HTML.

Current Progress: 68 raw product records; 35 cleaned usable products

Final Goal: 500 furniture products across 6–8 categories

Exploratory Data Analysis



Dataset Summary

35 products across multiple categories
Average price: \$325
Median price: \$249
Class balance:
17 high-interest / 18 low-interest

Key Findings

Most products fall within the \$50–\$300 price range
Product distribution varies across categories
Higher-priced products are less frequent

Data Quality

Small sample size (35 products)
Limited features in current dataset

Feature Engineering

Create binary target (high_interest)
Category encoding (one-hot)
Potential text features

Insights:

High_interest is defined based on review_count ranking within each category;
Mid-priced products (around \$100–\$300) are more likely to be high_interest;
Review counts are right-skewed — a few products dominate engagement

Proposed Modeling Approach

Target Variable: `high_interest = 1 if top 50% within each category` → Binary Classification

Baseline

Logistic Regression

Current baseline:

Feature = price only

Accuracy $\approx$ 64%

Cross-validation accuracy $\approx$ 49%

- price alone generalizes poorly

Metric: Accuracy

Primary Model

Random Forest

Handles non-linear relationships between price, category, and product attributes.
Built-in feature importance for explainability.

Metric: F1 + AUC

Stretch Goal

XGBoost

A stronger tabular model.

Capture non-linear effects

Compare feature importance with random forest

Metric: F1 + AUC

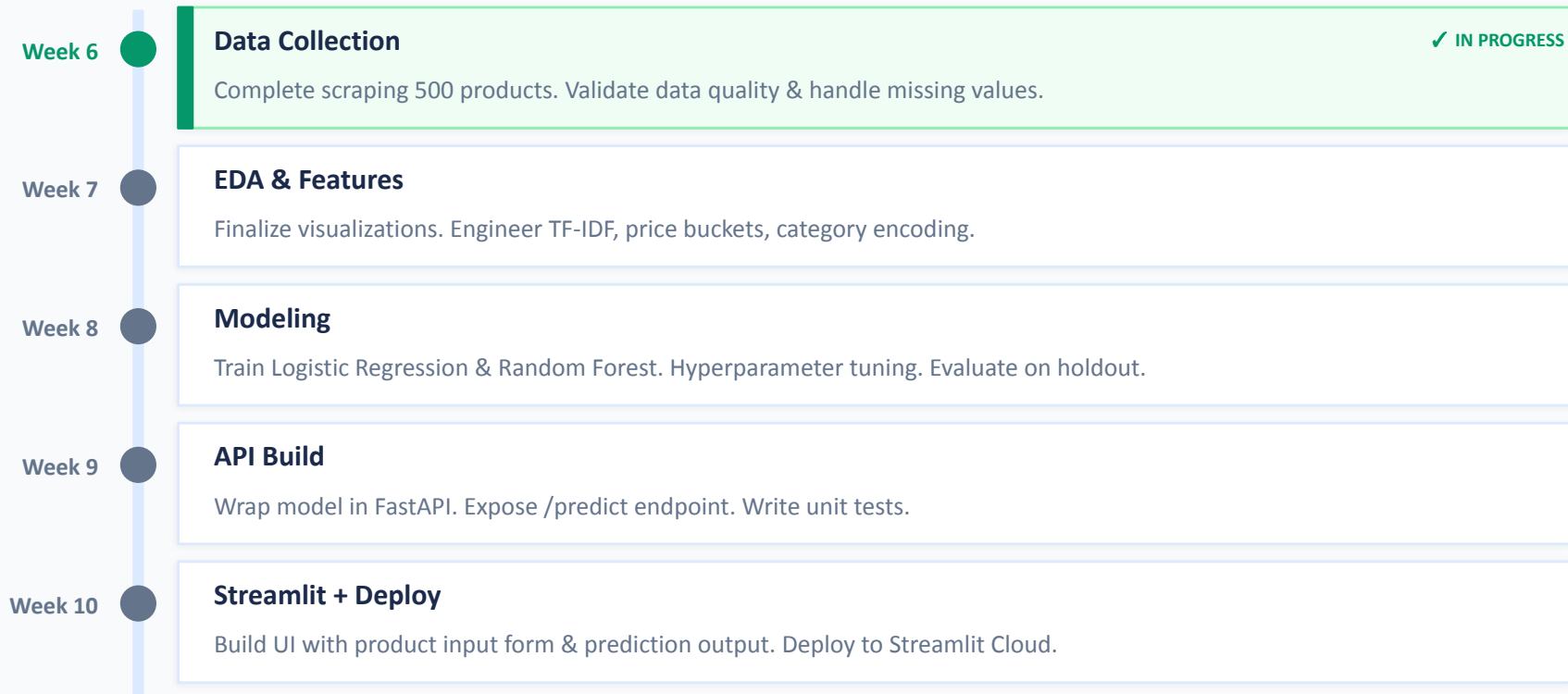
Input Features

<code>price</code>	— Numeric
<code>category</code>	— One-hot encoded
<code>description_length</code>	— Numeric

<code>material_keywords</code>	— TF-IDF (top 20)
<code>rating</code>	— Numeric
<code>color_count</code>	— Numeric
<code>price_bucket</code>	— Ordinal

Success Criteria: F1-score $\geq$ 0.7 after adding category/text features · clear feature importance analysis · optional Streamlit demo

Timeline & Next Steps



⚠ Risk: IKEA blocks scraper → Mitigation: Use cached data + IKEA open data API as fallback.

⚠ Risk: Insufficient reviews for binary split → Use quartile-based threshold instead of median.